

Answer: B

21 For a stationary wave in a closed pipe,

$$f_1 = \frac{v}{4L}, f_2 = \frac{3v}{4L}, f_3 = \frac{5v}{4L}$$

If $f_1 = 50\text{Hz}$,

$$f_2 = 3(50)\text{Hz} = 150\text{ Hz}$$

$$f_3 = 5(50)\text{ Hz} = 250\text{ Hz}$$

Answer: C

22 $d\sin\theta = n\lambda$